

K929 — PRE-REGISTERED audit criteria for the induced- a_2 / β -function computation, committed BLIND (before Elie’s #418 result). The key pre-commitment, and it reframes what “success” means: the SIGN of β_0 emerging from the geometry (ideally the SAME curvature as confinement) is the real win; “reproducing the standard 11/3” is a CONSISTENCY CHECK, not a derivation — because if BST’s emergent gauge sector is just standard 4D Yang-Mills, its a_2 is 11/3 *automatically* by Gilkey/Vassilevich bookkeeping, which discovers nothing about D_{IV}^5 . The FF-20 danger is dressing up a consistency check (or worse, an 11-weld) as a Tier-1 derivation. This note sets the bar before the computation, so the verdict can’t be retrofitted.

Keeper | 2026-07-26 Sun | Blind pre-registration for the strong-sector make-or-break (#418). Same discipline as K922/K924. The bar: does the SIGN come from the domain’s curvature, target-innocently — not from plugging N_c , N_f into the standard formula, and not from matching an “11.”

Standard-physics anchors (target-innocent, book-sourced — the reference, not BST)

- One-loop: $\beta_0 = 11/3 \cdot C_A - 4/3 \cdot T_F \cdot n_f$ ($= (11N_c - 2n_f)/3$ for $SU(N_c)$, Dirac fermions $T_F=1/2$). Asymptotic freedom $\Leftrightarrow \beta_0 > 0 \Leftrightarrow 11N_c > 2n_f$.
- Nielsen–Hughes decomposition (the physical origin — this is what BST must reproduce/route through): the gauge boson’s $-11/3 \cdot C_A$ = diamagnetic (orbital) +1/3 MINUS paramagnetic (spin/color-magnetic-moment) 4. The paramagnetic -4 is the ANTISCREENING = asymptotic freedom, and it is the gluon’s spin-1 color-magnetic moment (gyromagnetic $g=2$). So “the sign” physically = the paramagnetic spin term dominating the diamagnetic orbital term.
- $11/3 = 4 - 1/3$ is STANDARD heat-kernel/spectral-asymmetry math; it is NOT a BST integer.

THE TIERS (what counts as what — pre-committed)

- Tier-1 = the WIN. The SIGN $\beta_0 > 0$ emerges from the domain’s curvature, target-innocently. Specifically: the induced gauge-fluctuation a_2 ’s paramagnetic (color-magnetic-moment / antiscreening) term is produced by the non-abelian curvature of the hosted $SU(3)$ and dominates the diamagnetic term — derived from D_{IV}^5 ’s structure, NOT from evaluating the standard formula at $N_c=3$, $n_f=6$.
 - Tier-1★ (the deep win): S2 — the antiscreening curvature is the SAME object as the confinement mechanism (colored $\rightarrow$ zero Shilov overlap / Schur, K745). If the same non-abelian curvature that forces confinement forces the AF sign, confinement + AF are ONE geometric structure (as they are in QCD). THIS is the headline worth chasing.
- Tier-2 = a CONSISTENCY CHECK, not a discovery. “The induced a_2 reproduces standard $11/3 \cdot C_A - 2/3 \cdot n_f$.” Valuable only as: *BST’s emergent gauge sector IS standard 4D YM, so it does not BREAK QCD*. Do NOT present this as “we derived the 11” — if the operator is the standard YM fluctuation operator on emergent 4D, 11/3 falls out of standard math trivially. State it as “consistency: standard running embeds in D_{IV}^5 ,” never as a BST derivation of 11.
- Tier-3 = the BST-SPECIFIC PREDICTION. A genuine MODIFICATION — the induced operator DIFFERS from standard YM (via $N_{\max}$ -finiteness / the Bergman propagator / finite-sum structure), computed, giving a deviation from standard running (the $c_1=3/5$ territory). Only this is a new, falsifiable BST result on the coefficient.

PASS conditions (per tier)

- Tier-1 PASS: S1 — $\beta_0 > 0$ shown from the induced a_2 ’s paramagnetic term via the domain’s non-abelian curvature, with NO use of the observed $\beta_0=7$ or the standard 11/3 as an input to get the sign. (Target-

innocent = the sign is an output of the geometry.)

- Tier-1★ PASS: S2 — the antiscreening curvature operator is exhibited as the same object as the Schur/zero-Shilov confinement structure. (A real unification.)
- Tier-2 (consistency) PASS: the induced operator is shown to equal the standard YM fluctuation operator on emergent 4D, so its a_2 = standard, by book-sourced (Vassilevich) bookkeeping — stated as consistency, not derivation.
- Tier-3 PASS: a specific deviation term computed from a specific BST modification of the operator, falsifiable.

FIT / FAIL red flags (any → not a derivation)

- The “11” welded to $c_2(Q^5)=11$, $\dim K=11$, Weitzenböck $c_2=11$, $2n_C+1$, or $137=11^2+4^2$ — the FF-20 elevens. The 11/3 must be the Nielsen 4–1/3, not a costume of a BST 11.
- $\beta_0 = g = 7$ banked as a derivation — it is an identification ($7=g$ only at $n_f=6$, all flavors active; scale-dependent). Fine as “identification to upgrade”; NOT a derived coefficient.
- The sign or 11/3 obtained by plugging N_c, n_f into the standard formula and calling it derived (that only supplies BST’s integers to standard QFT — the current banked state, explicitly the thing to upgrade).
- The a_2 asserted without the actual emergent-4D + forced-chiral-content Seeley-DeWitt computation.
- Any heat-kernel constant fabricated rather than book-sourced (Vassilevich/Gilkey).

PRE-COMMITTED VERDICT CEILING (committed now)

- The area’s real deliverable is Tier-1 (the sign from geometry), and its best form Tier-1★ (unified with confinement) — I will sign “derived” only if S1 is target-innocent (sign is a geometric output), and “unified” only if S2 exhibits one operator.
- Tier-2 I will record as CONSISTENCY, never as “derived the 11.” A reproduced 11/3 from a standard operator is not a discovery about D_{IV}^5 .
- Tier-3 is the only route to a NEW coefficient claim and requires a computed deviation, not a story.
- My prior (stated so it can’t bias): the SIGN is achievable (it’s a mechanism/sign claim, BST’s strength, and the confinement curvature is right there); the coefficient is either a consistency check (likely) or a hard modification (unlikely to be clean). S1/S2 decide it, not the prior.

— Keeper K929, 2026-07-26. BLIND pre-registration, induced- a_2/β -function (#418). REFRAME: the SIGN $\beta_0>0$ from the domain’s curvature (target-innocent, ideally the SAME curvature as confinement = Tier-1★) is the WIN; “reproduce standard $11/3 \cdot C_A - 2/3 \cdot n_f$ ” is a CONSISTENCY CHECK (standard YM embeds in D_{IV}^5), NOT a derivation of “11” ($11/3 = \text{Nielsen } 4-1/3$, standard math). Tier-3 (a computed $N_{\max}/\text{Bergman MODIFICATION}$) is the only route to a new coefficient claim. Red flags: weld the 11 to $c_2/\dim K/\text{Weitzenböck}$ (FF-20 elevens); bank $\beta_0=g=7$ as derived; plug N_c, n_f into the standard formula and call it derived; assert a_2 without the emergent-4D+chiral computation; fabricate heat-kernel constants. Sign PASS = S1 ($\beta_0>0$ from curvature, no observed- $\beta_0/\text{standard-}11/3$ as input); unification PASS = S2 (antiscreening curvature = the Schur/zero-Shilov confinement operator). See [[Keeper_K927_STRONG_SECTOR_DYNAMICS_area_charter]], [[Keeper_K924_PRE_REGISTERED_FK_Wyler_3x3_audit_criteria]], [[feedback_commit_the_checker_half_blind]].